2 **(a)** **(i)** Define gravitational field at a point.

....

.... [1]

(ii) Figure 2.1 shows a horizontal section through a small piece of ground at the Earth's surface and the air at a small distance above it.

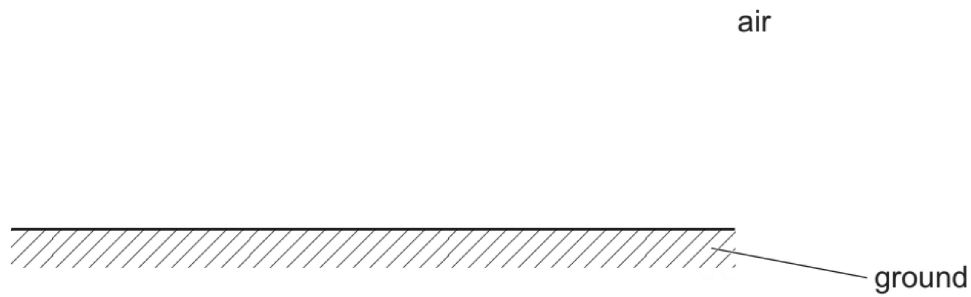

Figure 2.1

On Figure 2.1, draw four field lines to represent the gravitational field of the Earth in the air above the ground. [2]

(b) The Earth may be considered to be an isolated uniform sphere of radius 6.37×10^6 m.

(i) By considering the gravitational field strength at the Earth's surface, calculate a value for the mass of the Earth. Give your answer to three significant figures.

mass = ... kg [2]

(ii) Determine, to three significant figures, the gravitational potential at the Earth's surface. Give a unit with your answer.

gravitational potential = ... unit .... [2]

(iii) A satellite of mass 96.0 kg is in circular orbit at a height of 1.93×10^6 m above the Earth's surface.

Determine the gravitational potential energy of the satellite.

gravitational potential energy = ... J [2]

[Total: 9]